

$e = (1+dx)^{(1/dx)}$ can be easily derived from the definition of the function $f = e^x$ as the one having the property $df = f dx$

$$\begin{aligned}
 d(e^x) &= e^x dx \\
 \Rightarrow e^{(x+dx)} - e^x &= e^x dx \\
 \Rightarrow e^x e^{dx} - e^x &= e^x dx \\
 \Rightarrow e^x (e^{dx} - 1) &= e^x dx \\
 \Rightarrow e^{dx} - 1 &= dx \\
 \Rightarrow e^{dx} &= 1 + dx \\
 \Rightarrow \ln(e^{dx}) &= \ln(1 + dx) \\
 \Rightarrow dx &= \ln(1 + dx) \\
 \Rightarrow 1 &= (1/dx) \ln(1 + dx) \\
 \Rightarrow 1 &= \ln(1 + dx)^{(1/dx)} \\
 \Rightarrow e = e^1 &= (1 + dx)^{(1/dx)}
 \end{aligned}$$

where dx is an invertible infinitesimal ($dx \neq 0$ in the “classical” sense, and $dx \cdot \frac{1}{dx} = \frac{dx}{dx} = 1$) since the presented derivation doesn’t rely on nilpotency at all.

If in doubt, the function $\ln(x)$ is defined as the inverse function of e^x

$$\begin{aligned}
 \ln(e^x) &= e^{\ln(x)} = x, \\
 y = \ln(x) &\Leftrightarrow x = e^y
 \end{aligned}$$

with the differential

$$\begin{aligned}
 dx &= d(e^y) = e^y dy, \\
 dy &= dx/e^y, \\
 d \ln(x) &= dx/x
 \end{aligned}$$

and for the product $\ln(uv)$ it is

$$\begin{aligned}
 d \ln(uv) &= \frac{d(uv)}{uv} = \frac{(du)v + u(dv)}{uv} = \frac{du}{u} + \frac{dv}{v} \\
 &= d \ln(u) + d \ln(v) = d(\ln(u) + \ln(v)),
 \end{aligned}$$

then by the linearity of differentiation

$$d(\ln(uv) - (\ln(u) + \ln(v))) = 0,$$

thus $\ln(uv) - (\ln(u) + \ln(v)) = c$, where c is a constant, and since $e^0 = 1 \Leftrightarrow 0 = \ln(1)$ either $u = 1$ or $v = 1$ (or both) gives $c = 0$, therefore

$$\ln(uv) = \ln(u) + \ln(v)$$

Having that and placing $u = e^x$, $v = e^y$,

$$\begin{aligned}\ln(e^x e^y) &= \ln(e^x) + \ln(e^y) = x + y, \\ e^{\ln(e^x e^y)} &= e^{(x+y)}\end{aligned}$$

and finally

$$e^x e^y = e^{(x+y)}$$

proves the property used as $e^{(x+dx)} = e^x e^{dx}$.

$e^x = \sum_{i=0}^{\infty} \frac{x^i}{i!}$ I know as the proof of existence of the function $f = e^x$ by the definition $d(e^x) = e^x dx$. Presenting an example of such a function proves its existence. The function e^x is represented as the infinite sum of addends a_i

$$e^x = a_0 + a_1 + a_2 + a_3 + \dots$$

whose differential is

$$df = d(a_0 + a_1 + a_2 + a_3 + \dots) = da_0 + da_1 + da_2 + da_3 + \dots$$

Let $da_0 = 0$ so a_0 is constant and is equal to 1 since $e^0 = 1$, and for the subsequent addends

$$da_{(i+1)} = a_i dx,$$

then there will be $df = f dx$, as is needed.

It is known (if not, I can derive it as well) that $d(x^{(i+1)}) = (i+1)x^i dx$. Therefore let the common addend be $a_i = \eta_i x^i$ where each η_i is constant (and for $a_0 = \eta_0 x^0 = 1 \Rightarrow \eta_0 = 1$), then $da_{(i+1)} = a_i dx$ gives

$$\begin{aligned}\eta_{(i+1)} d(x^{(i+1)}) &= \eta_i x^i dx \\ \Rightarrow \eta_{(i+1)} (i+1) x^i dx &= \eta_i x^i dx \\ \Rightarrow \eta_{(i+1)} &= \frac{\eta_i}{i+1}\end{aligned}$$

Thus $\eta_1 = \frac{\eta_0}{0+1} = 1$, $\eta_2 = \frac{\eta_1}{1+1} = \frac{1}{2}$, $\eta_3 = \frac{\eta_2}{2+1} = \frac{1}{2 \cdot 3}$, $\eta_4 = \frac{\eta_3}{3+1} = \frac{1}{2 \cdot 3 \cdot 4}$, $\dots$ and in common

$$\begin{aligned}\eta_i &= \frac{1}{i!}, \quad a_i = \frac{x^i}{i!}, \\ e^x &= \sum_{i=0}^{\infty} a_i = \sum_{i=0}^{\infty} \frac{x^i}{i!}\end{aligned}$$

Checking the differential

$$\begin{aligned}
 d\left(\sum_{i=0}^{\infty} \frac{x^i}{i!}\right) &= d(1) + d\left(\sum_{i=1}^{\infty} \frac{x^i}{i!}\right) \\
 &= 0 + \sum_{i=1}^{\infty} d\left(\frac{x^i}{i!}\right) = \sum_{i=1}^{\infty} \frac{d(x^i)}{i!} \\
 &= \sum_{i=1}^{\infty} \frac{ix^{(i-1)}dx}{i!} = \sum_{i=1}^{\infty} \frac{x^{(i-1)}}{(i-1)!}dx = \sum_{i=0}^{\infty} \frac{x^i}{i!}dx
 \end{aligned}$$

Hence the function with the defining property $df = fdx$ really exists.

Assume there is also another representation $e^x = \varphi(x)$ for which $d\varphi(x) = \varphi(x)dx = e^x dx$ as well. Then the differential of the difference

$$d\left(\varphi(x) - \sum_{i=0}^{\infty} \frac{x^i}{i!}\right) = d\varphi(x) - d\left(\sum_{i=0}^{\infty} \frac{x^i}{i!}\right) = e^x dx - e^x dx = 0$$

and therefore $\varphi(x) = c + \sum_{i=0}^{\infty} \frac{x^i}{i!}$. Because $\varphi(x) = e^x$ for all x , $c = e^x - e^x = 0$ and $\varphi(x) = \sum_{i=0}^{\infty} \frac{x^i}{i!}$. Thus the representation $e^x = \sum_{i=0}^{\infty} \frac{x^i}{i!}$ is unique.